



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

**TECHNICAL SCIENCES P2
TEGNIJSE WETENSKAPPE V2**

2022

MARKING GUIDELINES/NASIEENRIGLYNE

MARKS/PUNTE: 75

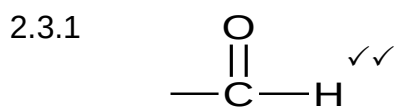
**These marking guidelines consist of 10 pages.
*Hierdie nasienriglyne bestaan uit 10 bladsye.***

QUESTION/VRAAG 1

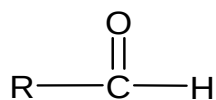
- 1.1 C ✓✓ (2)
- 1.2 B/C ✓✓ (2)
- 1.3 D ✓✓ (2)
- 1.4 B ✓✓ (2)
- 1.5 A ✓✓ (2)
- [10]**

QUESTION/VRAAG 2

- 2.1.1 E ✓ (1)
- 2.1.2 F ✓ (1)
- 2.2 C_nH_{2n-2} ✓✓ (2)

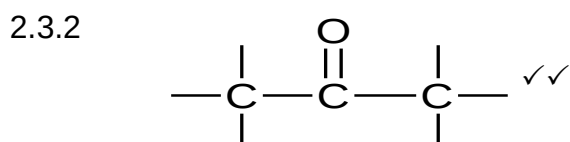


OR/OF



Marking criteria/Nasienkriteria:

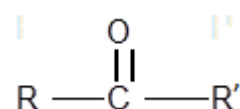
- If a bond is missing (0/2)
- Indien 'n binding uitgelaat is (0/2)



Marking criteria/Nasienkriteria:

- If a bond is missing (0/2)
- Indien 'n binding uitgelaat is (0/2)

OR/OF



(2)

2.4.1 Propan✓-2-ol✓

OR/OF

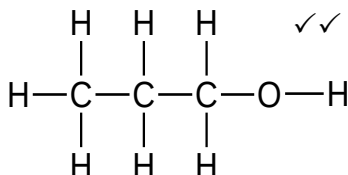
2 – propanol

Marking criteria/Nasienkriteria:

- Correct functional group and the stem
- Correct position of the functional group
- If a hyphen is missing ½
- Korrekte funksionele groep en die stam
- Korrekte posisie van die funksionele groep
- Indien koppelteken uitgelaat is ½

(2)

2.4.2



✓✓

Marking criteria/Nasienkriteria:

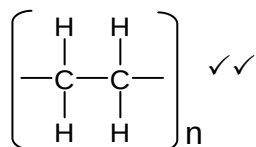
- Correct position of the functional group
- The whole structure correct.
- If a bond or hydrogen is missing ½
- Korrekte posisie van die funksionele groep
- Die hele struktuur is korrek.
- Indien koppelteken uitgelaat is ½

(2)

2.5.1 A large molecule composed of smaller monomer units ✓ covalently bonded to each other in a repeating pattern. ✓'n Groot molekule bestaande uit kleiner monomeereenhede, kovalent met mekaar verbind in 'n herhalende patroon.

(2)

2.5.2



✓✓

(2)

2.5.3 Polythene/Polyethylene ✓✓
Politeen/Poliëtileen

(2)

[18]

QUESTION/VRAAG 3

3.1.1 What is the relationship between chain length/molecular mass/surface area and boiling point in alkanes? ✓✓

Wat is die verhouding tussen kettinglengte/molekulêre massa/oppervlakarea en kookpunt in alkane?

OR/OF

How does the chain length/molecular mass/surface area affect the boiling point of alkanes?

Hoe beïnvloed die kettinglengte/molekulêre massa/oppervlakarea die kookpunt van alkane?

Marking criteria/Nasienkriteria:

- Dependant and independent variables correctly identified.
- Question correctly/appropriately asked about the relationship between the dependent and independent variable.
- Do not penalise if 'alkanes' is omitted.
- *Afhanklike en onafhanklike veranderlikes korrek geïdentifiseer*
- *Vraag korrek/toepaslik gevra oor die verhouding tussen die afhanklike en onafhanklike veranderlike*
- *Moenie penaliseer indien 'alkane' uitgelaat is nie.*

(2)

3.1.2 Chain length/molecular mass/surface area/compounds✓

Kettinglengte/molekulêre massa/oppervlakarea/verbindinge

(1)

3.1.3 Boiling point ✓/Kookpunt

(1)

3.1.4 Homologous series ✓/Functional group

Homoloë reeks/Funksionele groep

Accept/Aanvaar: Type of intermolecular forces./*Tipe intermolekulêre kragte*

(1)

3.2.1 London/Dispersion/Induced dipole forces.✓/Londen-/Dispersie-/Geïnduseerde-dipool-kragte

(1)

3.2.2 Incorrect ✓/Verkeerd

(1)

3.2.3 Apply negative marking from QUESTION 3.2.2./

Pas negatiewe nasien vanaf VRAAG 3.2.2 toe.

- The chain length/molecular mass/surface area decreases from compound **C** (butane) to compound **A** (ethane). ✓
- The smaller the chain length/molecular mass/surface area, the weaker the intermolecular forces. ✓
- The weaker the intermolecular forces, the lower is the boiling point. ✓
- *Die kettinglengte/molekulêre massa/oppervlakarea verminder van verbinding **C** (butaan) na verbinding **A** (etaan).*
- *Hoe kleiner die kettinglengte/molekulêre massa/oppervlakarea, hoe swakker die intermolekulêre kragte.*
- *Hoe swakker die intermolekulêre kragte, hoe laer is die kookpunt.*

OR/OF

- The chain length/molecular mass/surface area increases from compound **A** (ethane) to compound **C** (butane).
- The larger the chain length/molecular mass/surface area, the stronger the intermolecular forces.
- The stronger the intermolecular forces, the higher is the boiling point.
- *Die kettinglengte/molekulêre massa/oppervlakarea vergroot vanaf verbinding **A** (etaan) na verbinding **C** (butaan).*
- *Hoe groter die kettinglengte/molekulêre massa/oppervlakarea, hoe sterker die intermolekulêre kragte.*
- *Hoe sterker die intermolekulêre kragte, hoe hoër is die kookpunt.* (3)

3.2.4 Boiling point increases with an increase in chain length/molecular mass/surface area. ✓✓ */Kookpunt styg met 'n toename in kettinglengte/molekulêre massa/oppervlakarea.*

OR/OF

Boiling point decreases with a decrease in chain length/molecular mass/surface area. */Kookpunt daal met 'n afname in kettinglengte/molekulêre massa/oppervlakarea.*

(2)
[12]

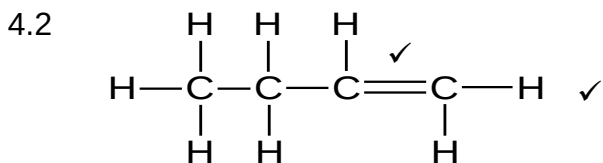
QUESTION/VRAAG 4

4.1.1 Addition/hydration (reaction) ✓
 Addisie/hidrasie (reaksie)

(1)

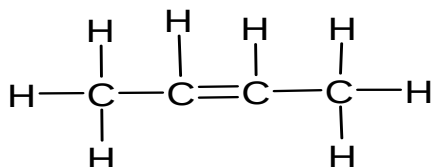
4.1.2 Substitution (reaction). ✓
 Substitusie (reaksie)

(1)



But - 1 - ene/1 - butene
 But - 1 - een/1 - buteen

OR/OF



But - 2 - ene/2- butene
 But - 2 - een/2-buteen

Marking criteria (Structure)/Nasienkriteria (Struktuur):

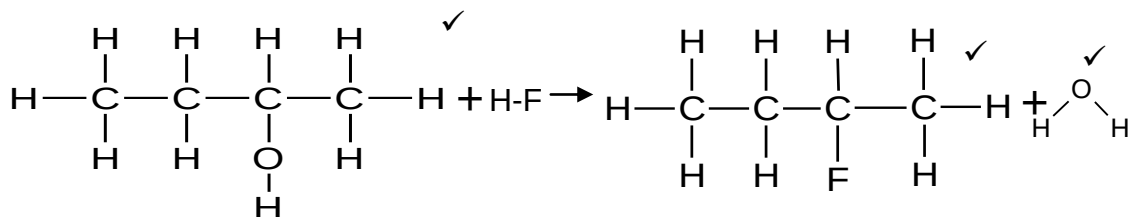
- Correct functional group
- The whole structure is correct.
- If a bond or hydrogen is missing ½
- Korrekte funksionele groep
- Die hele struktuur is korrek
- As verbinding of waterstof uitgelaat is ½

Marking criteria (IUPAC name)/Nasienkriteria (IUPAC-naam):

- Correct functional group and the stem
- Correct position of the functional group
- If a hyphen is missing ½
- Korrekte funksionele groep en die stam
- Korrekte posisie van die funksionele groep
- As 'n koppelteken uitgelaat is ½

(4)

4.3



Marking criteria/

Nasienkriteria:

- 2-florobutane 1 mark
- Water 1 mark
- Reactants 1 mark
- If a bond is missing, penalise 1 mark.
- 2-florobutaan 1 punt
- Water 1 punt
- Reaktanse 1 punt
- Indien 'n binding weggelaat is, penaliseer 1 punt.

(3)

4.4

- Excess water ✓/Oormaat water
- Acid catalyst/Suurkatalisator/H₂SO₄ (Sulphuric acid)/H₂SO₄ (Swawelsuur)/H₃PO₄ (Phosphoric acid)/H₃PO₄ (Fosforsuur) ✓

(2)

[11]

QUESTION/VRAAG 5

- 5.1.1 An electrode where oxidation takes place. ✓✓
'n Elektrode waar oksidasie plaasvind.

Marking criteria/Nasienkriteria:

- When 'electrode' is omitted: $\frac{1}{2}$
- If 'oxidation' is omitted: $\frac{0}{2}$
- Wanneer 'elektrode' weggelaat is: $\frac{1}{2}$
- As 'oksidasie' weggelaat is: $\frac{0}{2}$

(2)

- 5.1.2 The decomposition of a substance when an electric current is passed through it. ✓✓
Die opbreking (ontbinding) van 'n stof wanneer 'n elektriese stroom daardeur gaan.

ORIOF

The chemical process in which electrical energy is converted to chemical energy.

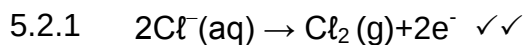
Die chemiese proses waar elektriese energie in chemiese energie omgeskakel word.

ORIOF

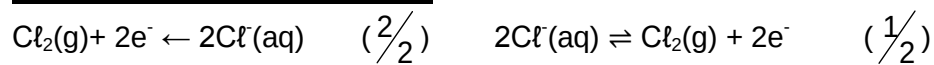
The use of electrical energy to produce a chemical change.

Die gebruik van elektriese energie om 'n chemiese verandering te veroorsaak.

(2)



Marking criteria/Nasienkriteria:



NOTE: Do not penalise if the phases are not included.

LET WEL: Moenie penaliseer as die fases nie ingesluit is nie.

(2)

5.2.2 Reduction ✓/Reduksie

(1)

5.3.1 Copper (II) ions ✓/Koper(II)ione

OR/OF



(1)

5.3.2 Chloride ions ✓/Chloriedione

OR/OF



(1)

5.4 Chlorine (gas) ✓/Chloor(gas)

(1)

[10]

QUESTION/VRAAG 6

6.1.1 Chemical (energy) to electrical (energy). ✓✓
Chemiese (energie) na elektriese (energie). (2)

6.1.2 • Temperature of 25 °C/298 K/temperatuur van 25 °C/298 K ✓
• Concentration of an electrolyte is 1 mol·dm⁻³ /
Konsentrasie van elektroliet is 1 mol·dm⁻³ ✓ (2)

6.2.1 $\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$ ✓✓

Marking criteria/Nasienkriteria:



NOTE: Do not penalise if the phases are not included.

LET WEL: Moenie penaliseer as die fases nie ingesluit is nie.

(2)

6.2.2 $\text{Cu}(\text{s}) \longrightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$ ✓✓

Marking criteria/Nasienkriteria:



NOTE: Do not penalise if the phases are not included.

LET WEL: Moenie penaliseer as die fases nie ingesluit is nie.

(2)

6.3 Apply positive marking from QUESTIONS 6.2.1 and 6.2.2./
Pas positiewe nasien vanaf VRAAG 6.2.1 en 6.2.2 toe.

OPTION/OPSIE 1	OPTION/OPSIE 2
$E_{\text{cell/sel}}^{\theta} = E_{\text{cathode/katode}}^{\theta} - E_{\text{anode}}^{\theta}$ ✓ $= 0,80 \text{ ✓} - (+0,34) \text{ ✓}$ $= 0,46 \text{ V ✓}$	$\text{Cu} \longrightarrow \text{Cu}^{2+} + 2\text{e}^- \quad -(+0,34) \text{ ✓}$ $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag} \quad (0,80) \text{ ✓}$ $\text{Ag} + \text{Cu}^{2+} \longrightarrow \text{Ag}^+ + \text{Cu}^{2+} \text{ ✓} \quad (0,46 \text{ V}) \text{ ✓}$

The cell is spontaneous. ✓/Die sel is spontaan.

NOTE: Penalise if unconventional abbreviations are used.

LET WEL: Penaliseer as onkonvensionele afkortings gebruik is.

(5)

6.4 Apply positive marking from QUESTION 6.3./
Pas positiewe nasien vanaf VRAAG 6.3 toe.

(Calculated value of) emf/ E_{cell}^{θ} is positive. ✓
(Berekende waarde van) emk/ E_{sel}^{θ} is positief.

(1)
[14]

TOTAL/TOTAAL: 75